Preference Enhancement Engine Evidence companion

The full specimens behind the presentation: actual FiveK renditions, newly executed Gen 1 checkpoint outputs, and the team's recorded research figures.

FiveK 0412 · one original, five expert renditions



Original · a0412-_DGW6297



Expert A



Expert B



Expert C



Expert D



Expert E

Original DNG and expert TIFFs from MIT–Adobe FiveK. Bychkovsky, Paris, Chan & Durand, CVPR 2011. These expert images are dataset targets.

Actual Gen 1 checkpoint executions · CPU · 256 × 256

Control	RGBC condition	Output range	Mean absolute edit
Zero	0, 0, 0, 0	0–0.911	0.027539
Warm	+0.10, 0, −0.10, 0	0–0.991	0.035392
Cool	−0.10, 0, +0.10, 0	0–0.816	0.066946
Expert D reference	+0.110, +0.159, +0.099, +0.172	0–1	0.146042

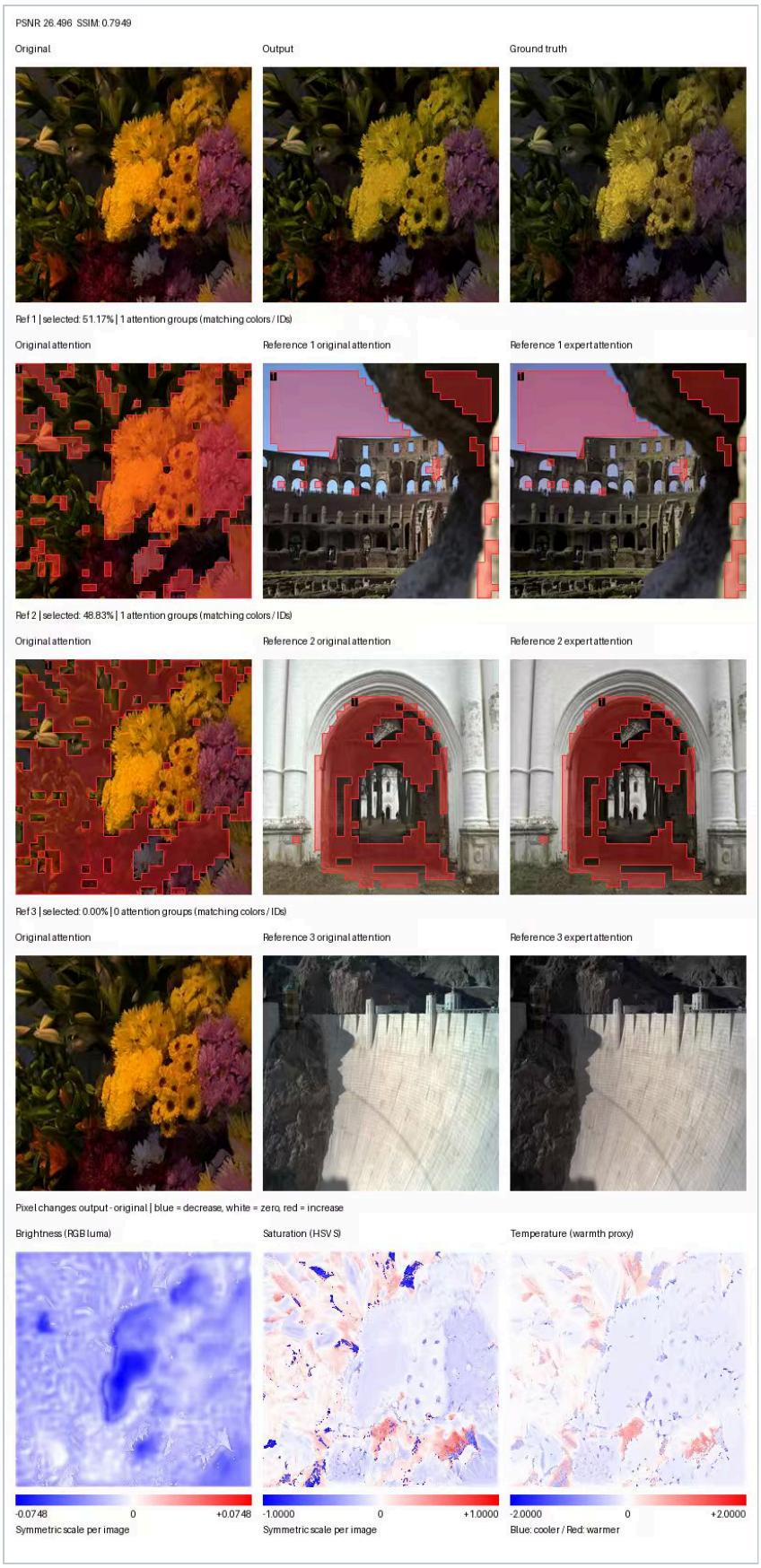
All four runs loaded the supplied weights strictly and produced finite outputs. The reference run uses same-scene Expert D with adapter strength 0.7. It illustrates an oracle control, without measuring unrelated-image transfer.

Processing matches the project loader: camera white balance for DNG, 8-bit RGB, Pillow TIFF conversion, 256-square model input. No additional ICC conversion was applied. The TIFF colour profiles and DNG rendering need standardisation before a colour-managed reproduction.

Flowers · full recorded diagnostic

PSNR 26.496 dB · SSIM 0.7949. Reference selection shares: 51.17%, 48.83%, 0%.

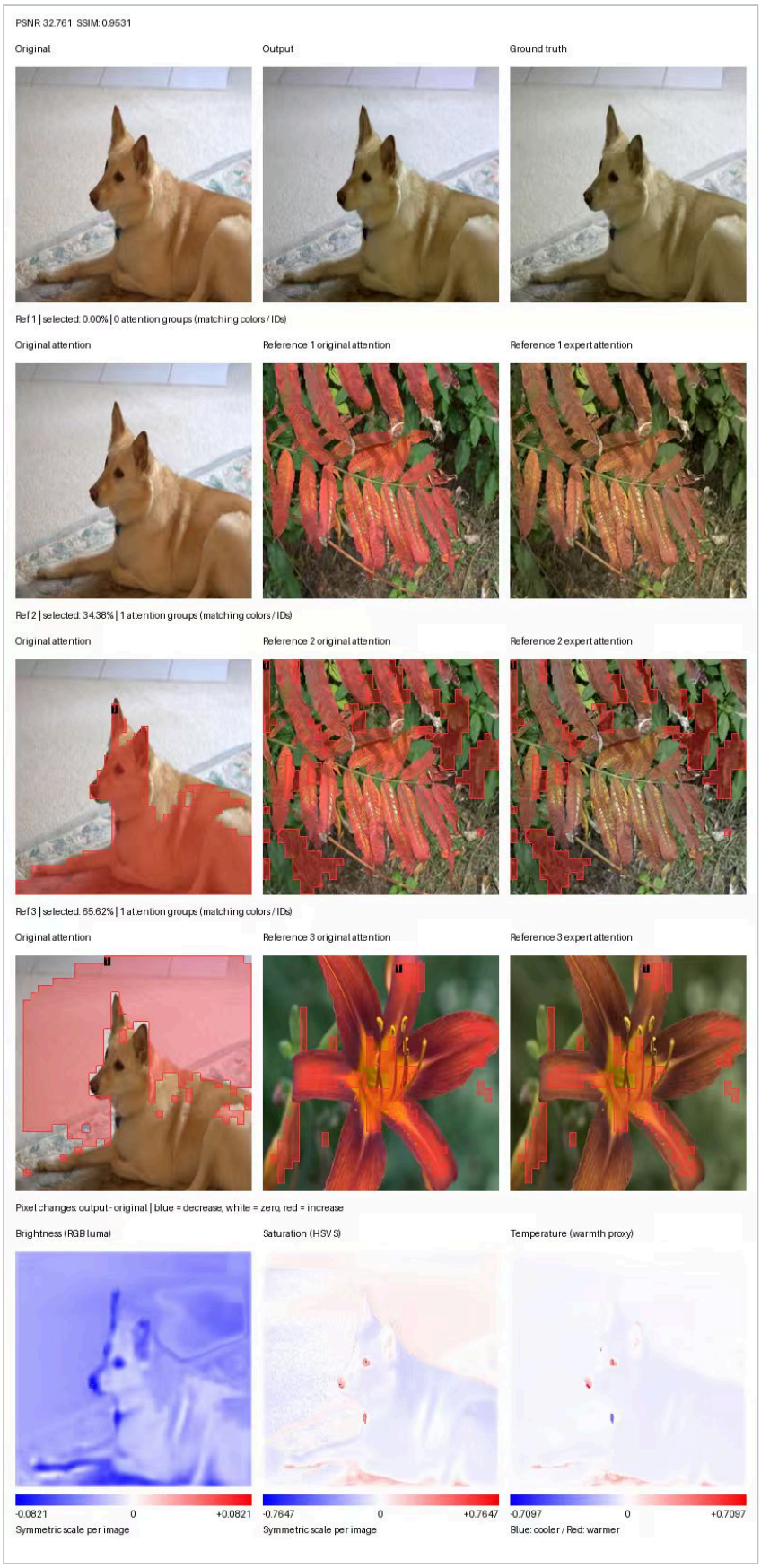
Shares count query regions selecting a reference. Matching colours and IDs show connected groups of strong attention links. The full supplied figure is unchanged.



Dog · full recorded diagnostic

PSNR 32.761 dB · SSIM 0.9531. Reference selection shares: 0%, 34.38%, 65.62%.

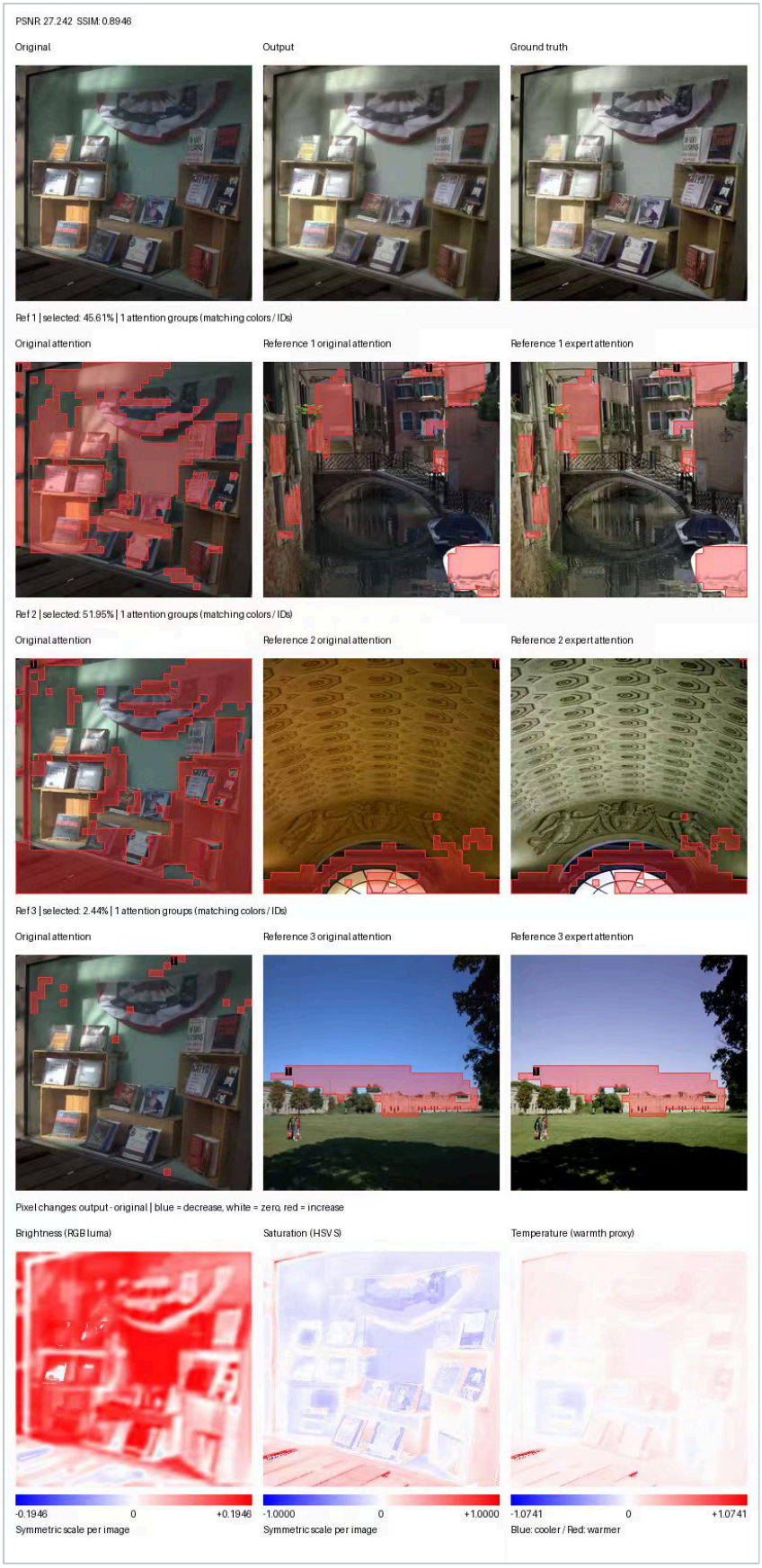
Two references are the same leaf scene with different edits; the third is a lily. This observation motivates testing correspondence across scene categories.



Books · full recorded diagnostic

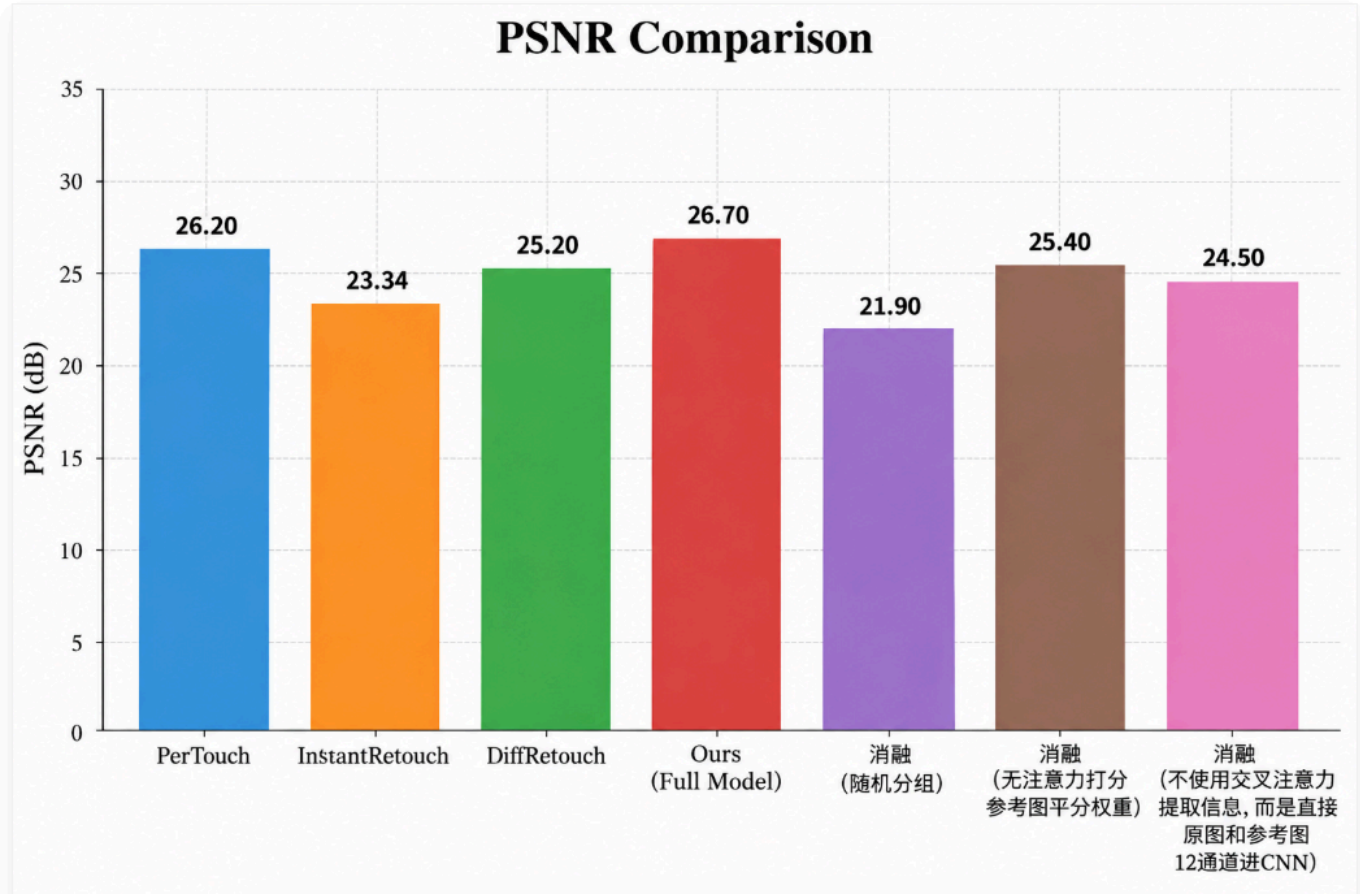
PSNR 27.242 dB · SSIM 0.8946. Reference selection shares: 45.61%, 51.95%, 2.44%.

The recorded checkpoint and raw attention tensors are unavailable. Metrics are transcribed from the original figure, with no recalculation from compressed images.



Reported comparison · original chart

The core deck focuses on the team's three ablations. The original chart is retained here for inspection.



Within-project results
Full: 26.70 dB. Random references: 21.90. Equal weights: 25.40. CNN fusion: 24.50.

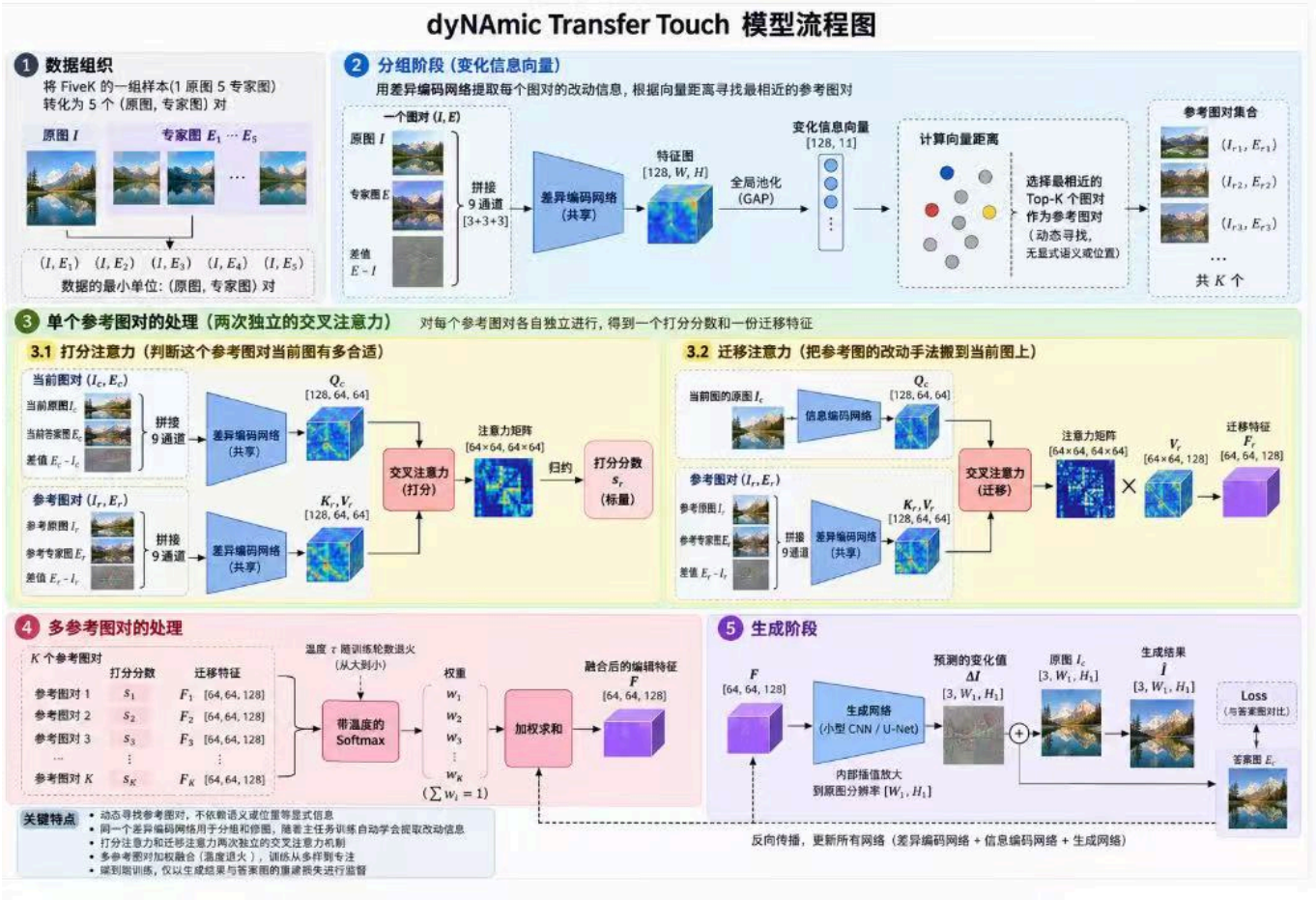
Comparison scope
Matching training splits, run configurations and random-seed uncertainty are not documented alongside the chart. The external-method entries do not establish a controlled ranking.

Separate milestone
The earlier 26.6 dB / 0.88 SSIM report concerns a separate evaluation scope. It is not the average of the three example panels.

Reproduction requirement
Archive weights, source revision, candidate-pool policy, split IDs and per-image metrics together for the next comparison.

Architecture · historical diagram and current contract

The supplied diagram records the team's architecture explanation. The core slides use the currently inspected code where the implementation differs.



Paired-edit retrieval
Original, expert target and their difference form a nine-channel input. A pooled 64-dimensional descriptor retrieves three pairs and excludes the same original.

Attention contract
At 256 × 256 input, Q/K scoring uses a 32 × 32 grid. Value features retain 64 × 64 resolution. Each query region chooses one reference; spatial correspondence inside a reference remains soft.

Training objective
Supervised RGB L1 plus a 0.001 residual penalty. Candidate embeddings refresh per training chunk. This code does not implement a reinforcement-learning objective.

Deployment boundary
Query expert information still enters retrieval. A persistent user-history store and query-only selection are future integration work. The current web demo uses the separate Gen 1 control network.